Unit – V

Cloud Computing (Course Code: 4360709)

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Unit– V

Security and Privacy Issues in Cloud Computing

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Unit– V Security and Privacy Issues in Cloud Computing

CO5 - e) To implement comprehensive security measures, ensuring the security and privacy of data

5.1 Conduct a security risk assessment for a cloud infrastructure, identifying vulnerabilities and proposing mitigation strategies.

5.2 Perform a privacy impact assessment for a cloud-based application, considering data life cycle and key privacy concerns.

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Infrastructure Security

- The Network level
- The Host level
- The Application Level

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Infrastructure Security

- It deals with the threats, risks, and challenges that are associated with the security of the organization's IT infrastructure such as the host, network, and application levels, this approach is commonly used by security practitioners whereas Non-IT security associates are advised not to equate the infrastructure security with access management's infrastructure as service security(IaaS).
- Besides that infrastructure security is more related to customers, as they have ramifications with threat, risk, and compliance management.

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Infrastructure Security at the Network Level

- There are no new attacks, vulnerabilities, or changes that need to be considered in this specific topology by the information security personnel, beside that our organization's IT infrastructure might be affected by the implementation of a private cloud but our current network topology probably will not get affected.
- whereas if we used the services of public clouds any changes in the security requirements will require a change in the network topology.
- Therefore, we must define some ways through which our existing network topology will interact with the topology of the cloud provider.

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Risk Factors Needed to be Addressed are

1. Integrity and Confidentiality of the In-Transit Data

2. Access Control Methods

3. Availability of the Services

4. Replace the Models Established in Network Zones and tiers within the Domains

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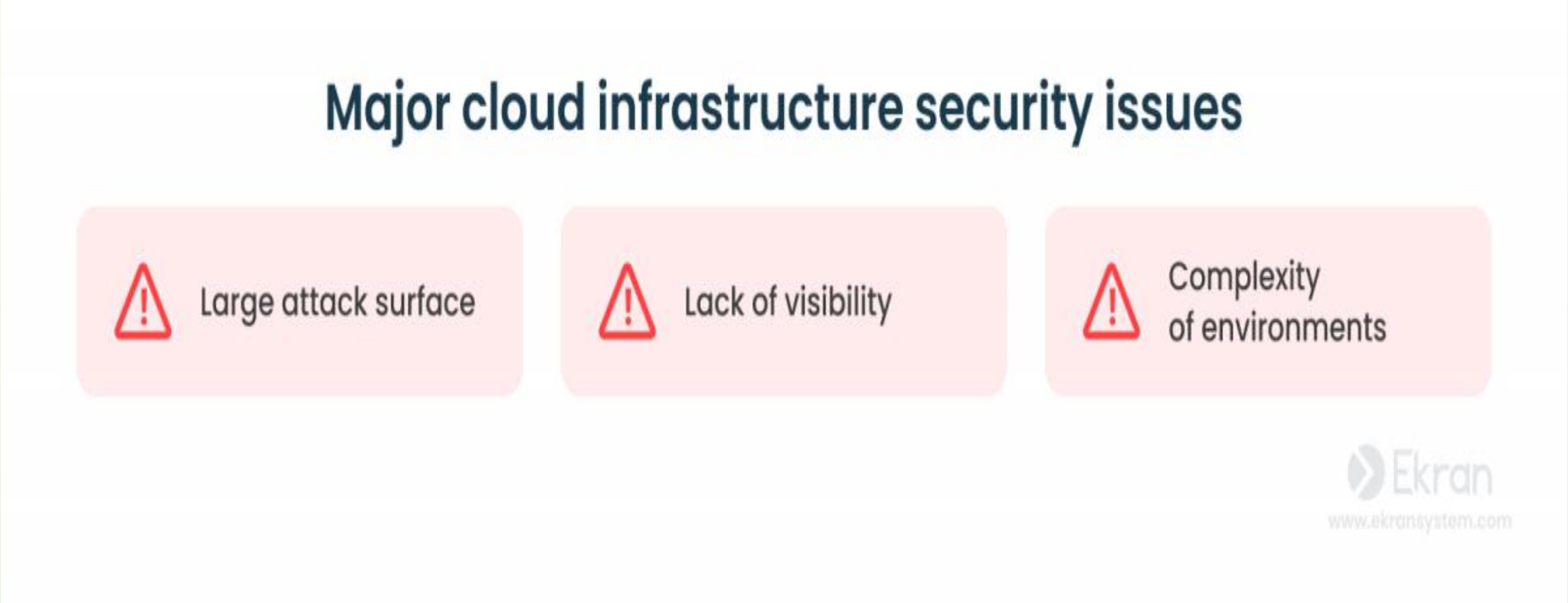
Major Cloud Infrastructure Security Issues



Major Cloud Infrastructure Security Issues

- **Large attack surface.** It can be challenging to clearly define the boundaries of a company's cloud environment. Systems and data might be attacked from many angles, including personal devices of remote employees, unauthorized third-party cloud applications and services, and public networks. Cloud data can be attacked both in rest and in transit.
- **Lack of visibility.** Some cloud providers possess complete control over cloud IT infrastructure without exposing that infrastructure to their customers. Organizations using cloud computing platforms may struggle to identify their cloud assets to take proper measures in order to efficiently protect their data. Additionally, it may not be easy to track the activity of your employees in the cloud.
- **Complexity of environments.** Certain organizations favor complex multi-cloud and hybrid environments due to their specific needs. This creates a problem of choosing the right cybersecurity tools that will operate both in the cloud and on-premises. Hybrid environments are challenging to implement and maintain and require a holistic approach to cybersecurity

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Benefits of Infrastructure Security

- Comprehensive Security for All Devices
- Easier to Scale
- Cost-Efficient
- Improved Disaster Recovery Processes
- Increased Innovation and Collaboration

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Cloud Infrastructure Best Practices

7 best practices for protecting cloud infrastructure

1. Secure access to the cloud
2. Manage user access privileges
3. Provide visibility with employee monitoring
4. Monitor privileged users
5. Educate employees against phishing
6. Ensure you meet IT compliance requirements
7. Respond to security incidents

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Cloud Infrastructure Best Practices

- 1. Secure access to the cloud Although most cloud providers have their own means of protecting their customers' infrastructure, you are still responsible for securing your organization's cloud user accounts and access to sensitive data. To reduce the risk of account compromise and credential theft, consider enhancing password management in your organization.

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Cloud Infrastructure Best Practices

- 2. Manage user access privileges To ensure employees can perform their duties efficiently, some organizations provide them with extensive access to systems and data at once. Accounts of such users are a goldmine for cyberattackers, as compromising them can make it easier to access critical cloud infrastructure and escalate privileges.

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- 3. Provide visibility with employee monitoring To increase transparency and secure the cloud infrastructure of your organization, you can use dedicated solutions to monitor your personnel's activity. By watching what your employees are doing during work hours, you'll be able to detect early signs of cloud account compromise or an insider threat.

Cloud Infrastructure Best Practices

- 4. Monitor privileged users: One of the key private cloud security best practices is keeping track of privileged users in your cloud infrastructure. Usually, system administrators and top management have more access to sensitive data than regular users. Consequently, privileged users can cause more damage to the cloud environment, whether maliciously or inadvertently.

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- 5. Educate employees against phishing: Monitoring user activity is not the only way to minimize the influence of the human factor inside your organization. To protect your cloud infrastructure even more, you can raise your personnel's cybersecurity awareness, with a particular emphasis on phishing.

Cloud Infrastructure Best Practices

6. Ensure you meet IT compliance requirements Cybersecurity compliance with standards, laws, and regulations aims to protect consumers' data and provide general guidance for organizations to better secure sensitive data. Without the right security controls and tools in your cloud infrastructure for IT compliance, your organization may lose millions of dollars in fines in case of a data breach.

7. Efficiently respond to security incidents Losses from a data breach can increase if you can't quickly detect, contain, and eradicate cybersecurity threats. The longer a threat remains in your cloud environment, the more data an attacker can exfiltrate or delete.

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Cloud Security Issues

- Cloud workloads and accounts being created outside of IT visibility (e.g., shadow IT)
- Incomplete control over who can access sensitive data
- Theft of data hosted in cloud infrastructure by malicious actor
- Lack of staff with the skills to secure cloud infrastructures
- Lack of visibility into what data is in the cloud
- Inability to prevent malicious insider theft or misuse of data
- Lack of consistent security controls over multi-cloud and on-premises environments
- Advanced threats and attacks against cloud infrastructure
- Inability to monitor cloud workload systems and applications for vulnerabilities
- Lateral spread of an attack from one cloud workload to another

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Host Security

- Host security describes how your server is set up for the following tasks:
- Preventing attacks.
- Minimizing the impact of a successful attack on the overall system
- Responding to attacks when they occur.

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Host Security

- It always helps to have software with no security holes. Good luck with that! In the real world, the best approach for preventing attacks is to assume your software has security holes.
- Each service you run on a host presents a distinct attack vector into the host.
- The more attack vectors, the more likely an attacker will find one with a security exploit. You must therefore minimize the different kinds of software running on a server.
- Given the assumption that your services are vulnerable, your most significant tool in preventing attackers from exploiting a vulnerability once it becomes known is the rapid rollout of security patches.
- Here’s where the dynamic nature of the cloud really alters what you can do from a security perspective. In a traditional data center, rolling out security patches across an entire infrastructure is time-consuming and risky.

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Host Security

- In the cloud, rolling out a patch across the infrastructure takes three simple steps:

- Patch your AMI (Amazon Machine image- used to create virtual servers in Amazon Elastic cloud) with the new security fixes.

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- Test the results.

- Relaunch your virtual servers.

Network Level Security

- All data on the network need to be secured. Strong network traffic encryption techniques such as Secure Socket Layer (SSL) and Transport Layer Security (TLS) can be used to prevent the leakage of sensitive information.

- Several key security elements such as data security, data integrity, authentication and authorization, data confidentiality, web application security, virtualization vulnerability, availability, backup, and data breaches should be carefully considered to keep the cloud up and running continuously.

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Application Level Security

- Studies indicate that most websites are secured at the network level while there may be security loopholes at the application level which may allow information access to unauthorized users.
- Software and hardware resources can be used to provide security to applications.
- In this way, attackers will not be able to get control over these applications and change them.
- XSS attacks, Cookie Poisoning, Hidden field manipulation, SQL injection attacks, DoS attacks, and Google Hacking are some examples of threats to application-level security which resulting from the unauthorized usage of the applications.

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Privacy Issue in Cloud

- Law: A system of rules are set for collection, use, storage and protection of personal data. They define rights of an individual and obligations for organizations to ensure data privacy and security.

- Access: To protect individuals from unauthorized access, privacy laws regulate access to personal data.

- Compliance: To protect individuals privacy, the organizations must adhere to established rules, regulations or standards.

- Retention: Privacy laws specify the amount of time for which an organization can retain data.

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- Audit: The organization requires to conduct regular audits to monitor compliance of privacy laws, identifying and addressing potential risks and breaches.

- Destruction: Organizations require to mandate reporting and addressing privacy breaches promptly with organizations taking measures to minimize harm harm and prevent future incidents.

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Data Security and Storage

- Data-in-transit
- Data-at-rest
- Processing of data, including multitenancy
- Data lineage
- Data provenance
- Data remanence

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Data Security Protection Layers, Risk Mitigation and Controls

Protection Layer	Risks Mitigated	Data Security Controls
Cloud SaaS/Service	**Lower levels (minus IaaS) +:** • Cloud/SaaS privileged access • Cloud environment compromise or failure • Remote legal access	**Encryption at Cloud SaaS/Service +** • **BYOK to cloud** • **Cloud encryption key management**
Application/ Database	**Lower levels +:** • Inside DB: DB Admins, DB Users • Inside App: Application Admins and Users • External compromise of these accounts	• **Application encryption** • **Database column encryption** • **TDE Key Management** • **Tokenization** • **Data masking** • **Database Access Monitoring**
File System / Volume	**Lower level + system level threats:** • External threats/breach of system/ privileged accounts • Enterprise Privileged User/Insider • Cloud IaaS privileged user/breach	• **File level encryption and access controls** • **Privileged user management**
Disk or other Media	**Physical device level:** • Loss, theft or improper retirement of physical media	• **Full disk encryption** • **SAN/Storage array/other KMIP key management** • **Tape encryption** • **VM encryption**

Data storage security policies

- Enterprises should have written policies specifying the appropriate levels of security for the different types of data that it has.

- Obviously, public data needs far less security than restricted or confidential data, and the organization needs to have security models, procedures and tools in place to apply appropriate protections.

- The policies should also include details on the security measures that should be deployed on the storage devices used by the organization.

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- **Access control** — Role-based access control is a must-have for a secure data storage system, and in some cases, multi-factor authentication may be appropriate. Administrators should also be sure to change any default passwords on their storage devices and to enforce the use of strong passwords by users.
- **Encryption** — Data should be encrypted both while in transit and at rest in the storage systems. Storage administrators also need to have a secure key management systems for tracking their encryption keys.

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- **Data loss prevention** — Many experts say that encryption alone is not enough to provide full data security. They recommend that organizations also deploy data loss prevention (DLP) solutions that can help find and stop any attacks in progress.
- **Strong network security** — Storage systems don’t exist in a vacuum; they should be surrounded by strong network security systems, such as firewalls, anti-malware protection, security gateways, intrusion detection systems and possibly advanced analytics and machine learning based security solutions. These measures should prevent most cyberattacks from ever gaining access to the storage devices.

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- **Strong endpoint security** — Similarly, organizations also need to make sure that they have appropriate security measures in place on the PCs, smartphones and other devices that will be accessing the stored data. These endpoints, particularly mobile devices, can otherwise be a weak point in an organization's cyberdefenses.
- **Redundancy** — Redundant storage, including RAID technology, not only helps to improve availability and performance, in some cases, it can also help organizations mitigate security incidents.

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- **Backup and recovery** — Some successful malware or ransomware attacks compromise corporate networks so completely that the only way to recover is to restore from backups.
- Storage managers need to make sure that their backup systems and processes are adequate for these type of events, as well as for disaster recovery purposes.
- In addition, they need to make sure that backup systems have the same level of data security in place as primary systems.

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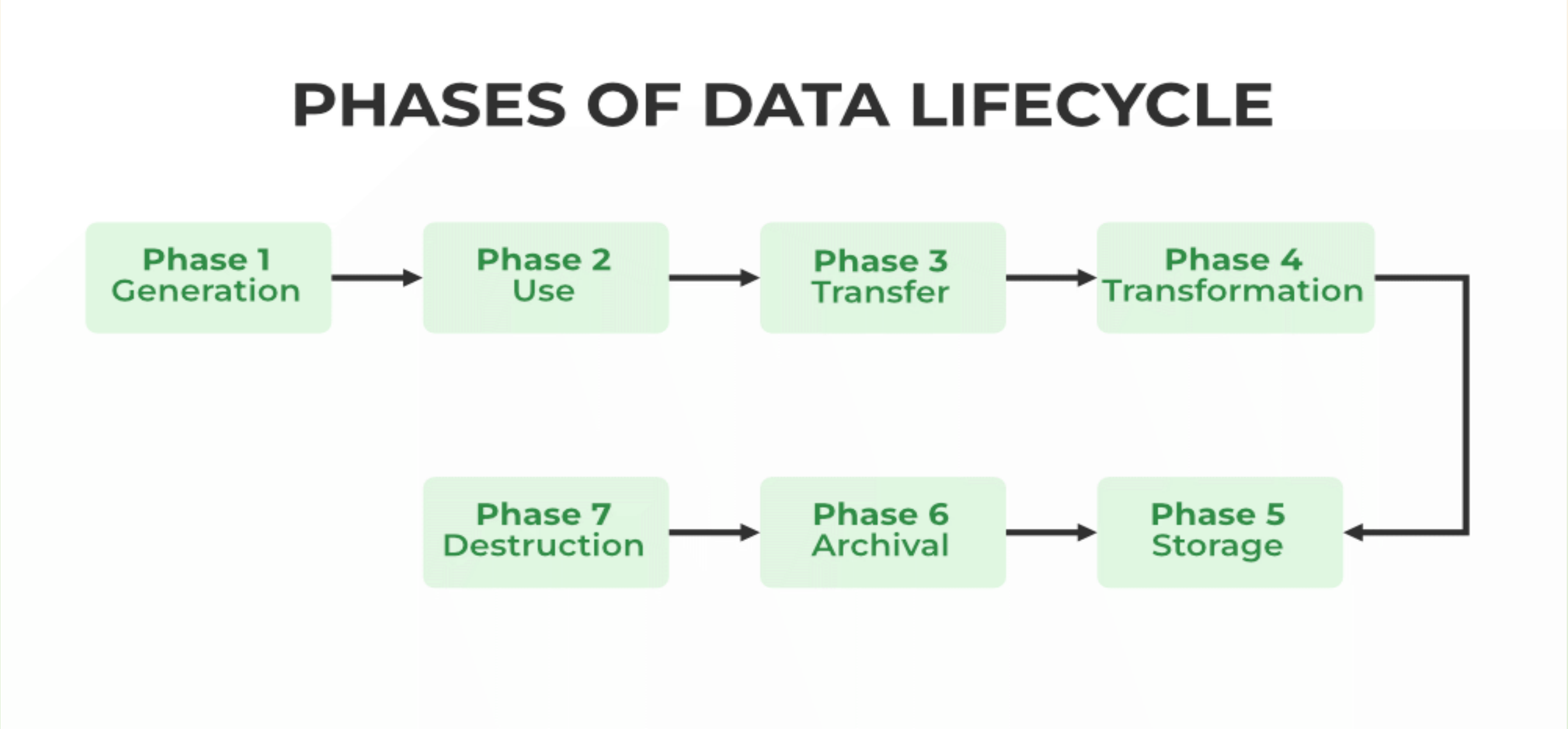
Data Life Cycle in Cloud



Components of Life Cycle

Generation of the Information

- **Ownership**: Who in the organization owns the user's data, and how is the ownership of data maintained within the organization?

- **Classification**: How and when is personally identifiable information classified? Are there any limitations on cloud computing on specific data cases?

- **Governance**: To ensure that personally identifiable information is managed and protected throughout its life-cycle

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Use of the Information

- **Internal v/s External**: Are personally identifiable information used only inside the organization or they are used outside the organization?
- **Third Party**: Is the personally identifiable information shared with third parties(organizations besides the parent company having data).
- **Appropriateness**: Is the personally identifiable information of users being correctly used for which it is intended?
- **Discovery/Subpoena:** Is the information stored in the cloud will enable the organization to comply with legal requirements in legal proceedings?

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Transfer of the Data

- **Public v/s Private Network**: Are the public networks secure(protected) enough while the personally identifiable information is transferred to the cloud?

- **Encryption Requirements:** Is the personally identifiable information encrypted while transmitted via a public network?

- **Access Control:** Appropriate access control measures should be taken on personally identifiable information when it is in the cloud.

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Transformation of Data

- **Derivation**:- While data is being transformed in the cloud, it should be protected and user limitations should be imposed on it.
- **Aggregations**:- The data should be aggregated so that we can ensure that it is no longer identifying any personal individual.
- **Integrity**:- Is the integrity of personally identifiable information maintained while it is in the cloud?

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Transformation of Data

Access Control:	• Appropriate access controls should be used on personally identifiable information while it is stored in the cloud so that only individuals with a need to know will be able to access it.
Structured v/s Unstructured:	• How the stored data will enable the organizations in accessing and managing the data in the future.
Integrity/Availability/Confidentiality:	• How data integrity, availability, and confidentiality are maintained in the cloud?
Encryption:	• The personally identifiable information should be encrypted while it is in the cloud.

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Archival

- **Legal and Compliance**: Personally identifiable information should have specific requirements that will instruct how long the data should be stored and archived.
- **Off-site Considerations**: Does the cloud service provider have the ability for long-term off-site storage and should also support the archival requirement?
- **Media Concerns**: Who will control the media and what is the organization's ability to recover in such cases when the media is lost?
- **Retention**: For how long the data should be retained on the cloud by the cloud service providers?

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Destruction of the Data

Secure:

- Does the cloud service providers destroy the personally identifiable information obtained by the customers to avoid a breach of information?

Complete:

- Does the personally identifiable information be completely destroyed? (erase the data, or it can be recovered)

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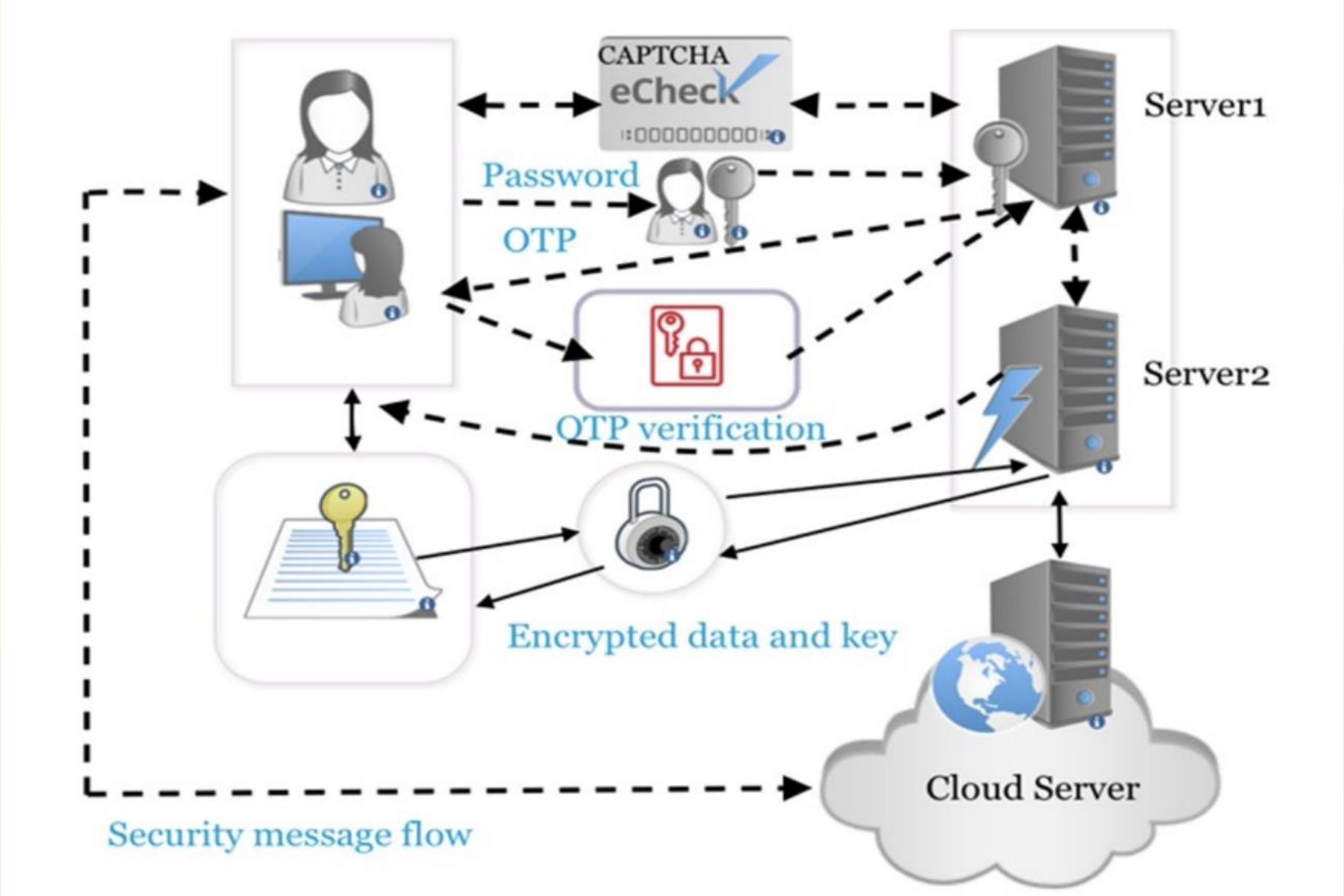
Key Privacy Concerns in Cloud



What Are the Key Privacy Concerns?

- Typically mix security and privacy
- Some considerations to be aware of:
  - Storage
  - Retention
  - Destruction
  - Auditing, monitoring and risk management
  - Privacy breaches
  - Who is responsible for protecting privacy?

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Storage

- Is it commingled with information from other organizations that use the same CSP?
- The aggregation of data raises new privacy issues
  - Some governments may decide to search through data without necessarily notifying the data owner, depending on where the data resides
- Whether the cloud provider itself has any right to see and access customer data?
- Some services today track user behaviour for a range of purposes, from sending targeted advertising to improving services

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Retention

- How long is personal information (that is transferred to the cloud) retained?
- Which retention policy governs the data?
- Does the organization own the data, or the CSP?
- Who enforces the retention policy in the cloud, and how are exceptions to this policy (such as litigation holds) managed?

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Destruction

- How does the cloud provider destroy PII at the end of the retention period?
- How do organizations ensure that their PII is destroyed by the CSP at the right point and is not available to other cloud users?
- Cloud storage providers usually replicate the data across multiple systems and sites—increased availability is one of the benefits they provide.
  - How do you know that the CSP didn't retain additional copies?
  - Did the CSP really destroy the data, or just make it inaccessible to the organization?
  - Is the CSP keeping the information longer than necessary so that it can mine the data for its own use?

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Auditing, monitoring and risk management

- How can organizations monitor their CSP and provide assurance to relevant stakeholders that privacy requirements are met when their PII is in the cloud?
- Are they regularly audited?
- What happens in the event of an incident?
- If business-critical processes are migrated to a cloud computing model, internal security processes need to evolve to allow multiple cloud providers to participate in those processes, as needed.
  - These include processes such as security monitoring, auditing, forensics, incident response, and business continuity

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Privacy breaches

- How do you know that a breach has occurred?
- How do you ensure that the CSP notifies you when a breach occurs?
- Who is responsible for managing the breach notification process (and costs associated with the process)?
- If contracts include liability for breaches resulting from negligence of the CSP?
  - How is the contract enforced?
  - How is it determined who is at fault?

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Who is responsible for protecting privacy?

- Data breaches have a cascading effect
- Full reliance on a third party to protect personal data?
- In-depth understanding of
- Organizations can tra
- Risk assessment and
  critical.
- Many new risks and unknowns
  - The overall complexity of privacy protection in the cloud represents a bigger challenge.

e.g., Suppose a hacker breaks into Cloud Provider A and steals data from Company X. Assume that the compromised server also contained data from Companies Y and Z.

- **Who investigates this crime?**
- **Is it the Cloud Provider, even though Company X may fear that the provider will try to absolve itself from responsibility?**
- **Is it Company X and, if so, does it have the right to see other data on that server, including logs that may show access to the data of Companies Y and Z?**

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Protecting Privacy

- Studies indicate that most websites are secured at the network level while there may be security loopholes at the application level which may allow information access to unauthorized users.
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THANK YOU

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